

# APMC Data Analysis Report

Source Analysis: APMC data/apmc.ipynb Date: 2026-02-16

## 1. Executive Summary

This report summarizes the analysis performed on the APMC Rajkot market data. The analysis focuses on commodity pricing, supply (arrival) trends, market volatility, and the relationship between supply and price.

## 2. Key Findings

### 2.1 Commodity Pricing Analysis

The analysis identified the commodities with the highest and lowest average rates (based on normalized data).

**Top 5 Most Expensive Commodities:**

1. **Rajkoo Bee** (Highest Avg Rate: ~0.57)
2. **Sesumum-Black** (Black Sesame)
3. **Cummin** (Jeera)
4. **Kalonji**
5. **Seasmum** (Sesame)

**Top 5 Least Expensive Commodities:**

1. **Cabbage** (Lowest Avg Rate: ~0.005)
2. **Papaya**
3. **Green Corn**
4. **Onion**
5. **Water Melon**

### 2.2 Price Volatility

We examined which commodities experience the most significant price fluctuations (standard deviation).

**Highest Price Volatility:**

1. **Rajkoo Bee** (Highest fluctuation)
2. **Green Garlic**
3. **Chillies**
4. **Lemon**
5. **Choli Shing**

These commodities carry higher market risk or seasonal sensitivity compared to others.

### 2.3 Supply vs. Price Relationship

A correlation analysis was conducted to determine if higher market arrivals lead to lower prices (standard supply/demand theory).

- **Correlation Coefficient:** -0.023
- **Conclusion:** There is a **negligible negative correlation** between arrival quantity and average price.
- **Interpretation:** The price of commodities in this market is likely driven by factors other than just immediate daily arrival volume (e.g., quality, external demand, seasonality, or global market trends). The scatter plot confirms this weak relationship.

## 2.4 Seasonal Trends

Monthly average price tracking was performed.

- **DF1 Trend:** Peak average rates observed around January (~0.18).
- **DF2 Trend:** Prices fluctuated between ~0.11 and ~0.14 normalized units throughout the year, with a slight dip around May and peaks in December/January.

## 3. Methodology Notes

- **Data Sources:** `apmc_rajkot_rates.csv` and `apmc_rajkot.csv`.
- **Preprocessing:**
  - Dates converted to datetime objects.
  - `Average Rate` calculated as mean of `lowrate` and `highrate`.
  - Data normalized using `MinMaxScaler` (values scaled between 0 and 1) for comparative analysis.